

Query Results

Question 1: How many entries in the database are from applicants who applied for Fall 2026?

Result: Fall 2026 applicant count: 33,207

Query:

```
"""
    SELECT COUNT (*) from applicants
    WHERE term = 'Fall 2026';
"""
```

Explanation: This query counts all rows in the applicants table where the application term is Fall 2026.

Question 2: Among entries that provide a nationality classification, what percentage are international students?

Result: Percent international: 31.67%

Query:

```
"""
    SELECT ROUND(
        100.0 * COUNT(*) /
        (
            SELECT COUNT(*)
            FROM applicants
            WHERE us_or_international IS NOT NULL
        ),
        2
    )
    FROM applicants
    WHERE us_or_international = 'International (my highest degree is from outside USA)'
    AND us_or_international IS NOT NULL;
"""
```

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Explanation: This query calculates the percentage of international applicants who report a nationality classification. The inner query counts all the entries with a reported classification, while the outer query counts those specifically classified as international. Multiplying this ratio by 100 and rounding to two decimal places produces the international-student percentage.

Question 3: What is the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Result: Average GPA: 3.77, Average GRE Quantitative: 261.49, Average GRE Verbal: 161.15, Average GRE Analytical Writing: 9.17

Query:

```
sql1 = """
    SELECT ROUND(AVG(gpa)::numeric, 2)
    FROM applicants;
    """
```

```
sql2 = """
    SELECT ROUND(AVG(gre)::numeric, 2)
    FROM applicants;
    """
```

```
sql3 = """
    SELECT ROUND(AVG(gre_v)::numeric, 2)
    FROM applicants;
    """
```

```
sql4 = """
    SELECT ROUND(AVG(gre_aw)::numeric, 2)
    FROM applicants;
    """
```

Explanation: These queries separately calculate the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores across applicants. It uses SQL's AVG() function, which excludes NULL values so each average is based only on applicants who provided

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their GPA. The datatype is converted to numeric from its base type of REAL in order to apply the ROUND function, and the results are rounded to two decimal places.

Question 4: What is the average GPA of American applicants who applied for Fall 2026?

Result: Average Fall 2026 American applicant GPA: 3.79

Query:

```
""""
SELECT ROUND(AVG(gpa)::numeric, 2)
FROM applicants
WHERE term = 'Fall 2026'
AND us_or_international = 'American'
AND gpa IS NOT NULL;
""""
```

Explanation: This query filters the database to applicants who are American, applied for Fall 2026, and reported a GPA. It then calculates the average of their GPA values using a similar methodology to the previous question, and rounds the result to two decimal places.

Question 5: What percentage of Fall 2025 entries are acceptances?

Result: Fall 2025 acceptance percentage: 40.73%

Query:

```
""""
SELECT ROUND(
    100.0 * COUNT(*) /
    (
        SELECT COUNT(*)
        FROM applicants
        WHERE term = 'Fall 2025'
    ),
    2
```

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```
)  
FROM applicants  
WHERE term = 'Fall 2025'  
AND status = 'Accepted';  
"""
```

Explanation: The outer query counts accepted Fall 2025 entries, while the inner query counts all Fall 2025 entries; dividing the accepted count by the total count and multiplying by 100 gives the acceptance percentage.

Question 6: What is the average GPA of accepted applicants who applied for Fall 2026?

Result: Average Fall 2026 accepted applicant GPA: 3.78

Query:

```
"""  
SELECT ROUND(AVG(gpa)::numeric, 2)  
FROM applicants  
WHERE term = 'Fall 2026'  
AND status = 'Accepted'  
AND gpa IS NOT NULL;  
"""
```

Explanation: This query filters for applicants who applied for Fall 2026, were Accepted, and reported a GPA. It then calculates the average of those GPA values (again, using a similar methodology to the prior questions).

Question 7: How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Result: JHU Computer Science masters applicants: 18

Query:

```
"""
```

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```
SELECT COUNT (*) from applicants
  WHERE university LIKE ANY (ARRAY['%Johns Hopkins University%', '%JMU%'])
  AND degree = 'Masters'
  AND program = 'Computer Science';
"""
```

Explanation: This query counts entries where the university name contains either “Johns Hopkins University” or “JMU”, the degree is Masters, and the program is Computer Science. It makes use of LIKE patterns to allow for matching of entries containing those university-name variations.

Question 8: How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at one of the following universities?

- **Georgetown University**
- **Massachusetts Institute of Technology / MIT**
- **Stanford University**
- **Carnegie Mellon University**

Result: Fall 2026 various university Computer Science PhD acceptances: 15

Query:

```
"""
SELECT COUNT (*) from applicants
  WHERE university LIKE ANY (ARRAY['%Georgetown University%', '%Massachusetts
Institute of Technology%', '%MIT%', '%Stanford University%', '%Carnegie Mellon
University%'])
  AND degree = 'PhD'
  AND program = 'Computer Science'
  AND status = 'Accepted'
  AND term = 'Fall 2026';
"""
```

Explanation: This query counts entries that meet all of the specified criteria: the university matches one of the four listed universities, the degree is PhD, the program is Computer Science, the applicant was Accepted, and the application term is Fall 2026. Because all conditions are combined with AND, only entries satisfying every requirement are counted.

Question 9: How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at one of the following universities (using LLM adjusted values)?

- **Georgetown University**
- **Massachusetts Institute of Technology / MIT**
- **Stanford University**
- **Carnegie Mellon University**

Result: Original-field count: 15, LLM-field count: 15, Difference: 0

Query:

```
sql1 = """
    SELECT COUNT (*) from applicants
        WHERE university LIKE ANY (ARRAY['%Georgetown University%', '%Massachusetts
Institute of Technology%', '%MIT%', '%Stanford University%', '%Carnegie Mellon
University%'])
        AND degree = 'PhD'
        AND program = 'Computer Science'
        AND status = 'Accepted'
        AND term = 'Fall 2026';
    """

sql2 = """
    SELECT COUNT (*) from applicants
        WHERE llm_generated_university LIKE ANY (ARRAY['%Georgetown University%',
'%Massachusetts Institute of Technology%', '%MIT%', '%Stanford University%', '%Carnegie
Mellon University%'])
        AND degree = 'PhD'
        AND llm_generated_program = 'Computer Science'
        AND status = 'Accepted'
        AND term = 'Fall 2026';
    """
```

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Explanation: The first query counts qualifying entries using the original university and program fields, while the second uses the LLM-adjusted university and program fields. Both queries apply the same degree, status, and term filters. My results for both the LLM fields and the original fields were the same, which I expected given that I tried really hard in the last assignment to ensure that the LLM didn't corrupt the original values.

Question 10: How many entries are applicants applying for a Physics PhD at West Virginia University?

Result: West Virginia University Physics PhD applicants: 11

Query:

```
"""
SELECT COUNT (*) from applicants
  WHERE university LIKE ANY (ARRAY['%West Virginia University%', '%WVU%'])
  AND program = 'Physics'
  AND degree = 'PhD'
"""
```

Explanation: This query counts applicants from West Virginia University (WVU) getting a PhD in Physics. It uses similar methodology to the prior questions; I chose this question because I did my undergrad at WVU originally and it seemed like I had a decent number of Physics applicants in my data set.

Question 11: What is the average GPA of applicants accepted to Johns Hopkins University for a master's degree who reported their GPA?

Result: JHU Masters acceptances average GPA: 3.69

Query:

```
"""
SELECT ROUND(AVG(gpa)::numeric, 2)
  FROM applicants
  WHERE university LIKE ANY (ARRAY['%Johns Hopkins University%', '%JMU%'])
"""
```

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```
AND status = 'Accepted'  
AND gpa IS NOT NULL  
AND degree = 'Masters';  
"""
```

Explanation: This query filters for applicants who were accepted to a master's program at Johns Hopkins University and who reported a GPA; it then calculates the average of the available GPA values.